Run report: item4-run-20261018T2049Z

Descriptive record of one recorded run of examples/genus4_res_census.rs (Parts T, H4, K) under `research/weil_descent_angles_20261018/PROTOCOL.md` (Addendum A and Amendments B–G and Addendum H, each committed before the run it governs). It lists what ran, on what host, and what the tool recorded, file by file. Interpretation is in the study's `REPORT.md`. Costs are counted operations (F_p multiplications and inversions, group operations, native operations), never wall-clock; the only times are the tool's own seconds. Isolation: none earned (a shared cloud container, or a RunPod CPU pod whose lscpu is in host.txt); this affects only the seconds, since every recorded value is an exact count or a counted operation total.

env.txt

```
run: item 4, 20261018T2049Z (Parts T, H4, K)
protocol commit: bdcf8f8ad (item 4 frozen); tool commit: 252b52998
rustc 1.97.0 (2d8144b78 2026-07-07)
host: Linux 6.18.44-fc-v139 x86_64, 4 cores, Intel(R) Xeon(R) Processor @ 2.10GHz, 15 GB
command: rustc --edition 2021 -O examples/genus4_res_census.rs -o g4 && G4_THREADS=4 ./g4 research/weil_descent_angles_20261018/results/item4-run-20261018T2049Z T H4 K
tool sha256: 8c6597ea8d5eb73604f8f2d3d724df4275deefce75db8944c6dc2c2b6edcc27b
threads: 4
smoke run before this run (not evidence): G4_LIMIT=200000 (4 s), Parts T and H4 as here; the K slice showed the expected orbit-outside-slice K5 failures only.
```

tool_log.txt

```
partT: n with an odd-t solution and n >= 3: [] (P5 predicts none)
partT: n = 2 odd-t cycle types: [6] [3, 3]
partT: n = 4 even-t solutions all in powers of 2: true
field F_3^12 ready 0.2s
H4: squarefree degree-10 binary forms = 104976; mass = 104976/48 = 2187 (check K1 expects 2187: ok)
H4: Res-shaped models = 1100; t histogram {-18: 18, -10: 96, -8: 144, -6: 96, -2: 192, 0: 204, 2: 180, 4: 48, 8: 48, 10: 48, 16: 24, 18: 2}
H4: check K2 (t even and cycle types in powers of 2) violations = 0
classify split: 36 classes, K5 violations 0
part K split: forms 43046720 smooth 21830400 mass 9475.0000 res_forms 81216 classes 36 (416.4s)
classify nonsplit: 39 classes, K5 violations 0
part K nonsplit: forms 43046720 smooth 27299520 mass 9479.0000 res_forms 110880 classes 39 (714.3s)
classify cone: 83 classes, K5 violations 0
part K cone: forms 14348906 smooth 8503056 mass 6561.0000 res_forms 101520 classes 83 (267.0s)
K: non-hyperelliptic mass total = 25515.000000 (check K3 literature value q^9+q^8-q^6 = 25515: match)
K: Res-shaped classes = 158; t histogram (class counts) {-18: 3, -17: 2, -16: 3, -15: 1, -14: 8, -13: 6, -12: 2, -11: 7, -10: 3, -9: 1, -8: 14, -7: 3, -6: 11, -5: 9, -4: 5, -3: 15, -2: 11, -1:
2, 0: 4, 1: 4, 2: 17, 3: 2, 4: 9, 5: 2, 7: 6, 9: 3, 10: 4, 11: 1}
K: check K4 violations = 0; check K5 violations = 0
K: P6 prime-order classes (82 - t prime, elliptic trace exists) = 9 : ["split:t=11:#E=71:aut=1", "nonsplit:t=-1:#E=83:aut=1", "cone:t=-7:#E=89:aut=1", "cone:t=9:#E=73:aut=3",
"cone:t=9:#E=73:aut=1", "cone:t=9:#E=73:aut=1", "cone:t=-1:#E=83:aut=1", "cone:t=-7:#E=89:aut=1", "cone:t=-7:#E=89:aut=1"]
total 1398.1s
```

checks.txt

```
partT: n with an odd-t solution and n >= 3: [] (P5 predicts none)
partT: n = 2 odd-t cycle types: [6] [3, 3]
partT: n = 4 even-t solutions all in powers of 2: true
H4: squarefree degree-10 binary forms = 104976; mass = 104976/48 = 2187 (check K1 expects 2187: ok)
H4: Res-shaped models = 1100; t histogram {-18: 18, -10: 96, -8: 144, -6: 96, -2: 192, 0: 204, 2: 180, 4: 48, 8: 48, 10: 48, 16: 24, 18: 2}
H4: check K2 (t even and cycle types in powers of 2) violations = 0
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2, 0: 4, 1: 4, 2: 17, 3: 2, 4: 9, 5: 2, 7: 6, 9: 3, 10: 4, 11: 1}
K: check K4 violations = 0; check K5 violations = 0
K: P6 prime-order classes (82 - t prime, elliptic trace exists) = 9 : ["split:t=11:#E=71:aut=1", "nonsplit:t=-1:#E=83:aut=1", "cone:t=-7:#E=89:aut=1", "cone:t=9:#E=73:aut=3",
"cone:t=9:#E=73:aut=1", "cone:t=9:#E=73:aut=1", "cone:t=-1:#E=83:aut=1", "cone:t=-7:#E=89:aut=1", "cone:t=-7:#E=89:aut=1"]
```

h4_res_shaped.csv (first 12 of 1100 rows)

code	a0..a10	N4	t	t_even	cycle_type	cycles_powers_of_2
19686	0 1 0 0 0 0 0 0 0 1 0	154	−18	true	1+1+4+4	true
19687	1 1 0 0 0 0 0 0 0 1 0	154	−18	true	1+1+4+4	true
19688	2 1 0 0 0 0 0 0 0 1 0	154	−18	true	1+1+4+4	true

19689	0 2 0 0 0 0 0 0 0 1 0	10	18	true	1+1+1+1+2+2+2	true
19716	0 2 0 1 0 0 0 0 0 1 0	82	0	true	1+1+8	true
19717	1 2 0 1 0 0 0 0 0 1 0	82	0	true	1+1+8	true
19718	2 2 0 1 0 0 0 0 0 1 0	82	0	true	1+1+8	true
19743	0 2 0 2 0 0 0 0 0 1 0	82	0	true	1+1+8	true
19744	1 2 0 2 0 0 0 0 0 1 0	82	0	true	1+1+8	true
19745	2 2 0 2 0 0 0 0 0 1 0	82	0	true	1+1+8	true
19932	0 2 0 0 0 1 0 0 0 1 0	18	16	true	1+1+8	true
19960	1 2 0 1 0 1 0 0 0 1 0	90	−2	true	1+1+8	true

k_classes.csv (first 12 of 158 rows)

type	rep_code	orbit_forms	aut	N4	t	numE	numE_prime	elliptic_trace_exists	form
split	708734	2304	1	94	−3	85	false	false	2*x^0y^0 + 1*x^0y^2 + 2*x^0y^3 + 1*x^1y^0 + 1*x^2y^3 + 1*x^3y^0
split	708761	2304	1	106	−6	88	false	false	2*x^0y^0 + 1*x^0y^2 + 2*x^1y^0 + 1*x^2y^3 + 1*x^3y^0
split	708989	2304	1	106	−6	88	false	false	2*x^0y^0 + 1*x^0y^1 + 2*x^0y^2 + 2*x^0y^3 + 1*x^1y^0 + 1*x^1y^1 + 1*x^2y^3 + 1*x^3y^0
split	728311	2304	1	114	−8	90	false	true	1*x^0y^0 + 1*x^0y^1 + 1*x^0y^2 + 1*x^0y^3 + 1*x^2y^1 + 1*x^2y^3 + 1*x^3y^0
split	729470	2304	1	90	−2	84	false	true	2*x^0y^0 + 1*x^0y^2 + 2*x^0y^3 + 2*x^1y^0 + 1*x^1y^1 + 1*x^1y^2 + 1*x^2y^1 + 1*x^2y^3 + 1*x^3y^0
split	729949	2304	1	102	−5	87	false	true	1*x^0y^0 + 1*x^0y^1 + 2*x^0y^3 + 2*x^1y^0 + 2*x^1y^2 + 1*x^2y^1 + 1*x^2y^3 + 1*x^3y^0
split	755479	2304	1	54	7	75	false	true	1*x^0y^0 + 2*x^0y^2 + 2*x^0y^3 + 2*x^1y^0 + 1*x^1y^2 + 1*x^2y^0 + 2*x^2y^1 + 1*x^2y^3 + 1*x^3y^0
split	768113	2304	1	78	1	81	false	true	2*x^0y^0 + 2*x^0y^1 + 1*x^0y^2 + 2*x^0y^3 + 2*x^1y^0 + 1*x^1y^1 + 1*x^2y^2 + 1*x^2y^3 + 1*x^3y^0
split	768506	2304	1	94	−3	85	false	false	2*x^0y^0 + 1*x^0y^1 + 2*x^0y^3 + 1*x^1y^0 + 1*x^1y^2 + 1*x^2y^2 + 1*x^2y^3 + 1*x^3y^0
split	769274	2304	1	66	4	78	false	true	2*x^0y^0 + 2*x^0y^1 + 1*x^0y^2 + 2*x^1y^0 + 2*x^1y^2 + 1*x^2y^2 + 1*x^2y^3 + 1*x^3y^0
split	769336	2304	1	94	−3	85	false	false	1*x^0y^0 + 2*x^0y^1 + 2*x^0y^2 + 2*x^0y^3 + 2*x^1y^0 + 2*x^1y^2 + 1*x^2y^2 + 1*x^2y^3 + 1*x^3y^0
split	774323	2304	1	38	11	71	true	true	2*x^0y^0 + 2*x^0y^1 + 1*x^0y^2 + 1*x^0y^3 + 1*x^1y^0 + 1*x^2y^0 + 1*x^2y^2 + 1*x^2y^3 + 1*x^3y^0

k_summary.csv (first 3 of 3 rows)

type	forms	smooth	mass_den	mass	smooth_with_N1_4	res_shaped_forms	res_shaped_classes	k5_violations	seconds
split	43046720	21830400	2304	9475.000000	4100256	81216	36	0	416.4
nonsplit	43046720	27299520	2880	9479.000000	5611680	110880	39	0	714.3
cone	14348906	8503056	1296	6561.000000	1925208	101520	83	0	267.0

part_t.csv (first 12 of 30 rows)

n	t_parity	solutions	cycle_types
2	odd	2	6 3+3
2	even	6	4+2 4+1+1 2+2+2 2+2+1+1 2+1+1+1+1 1+1+1+1+1+1
3	odd	0	
3	even	4	6+2 6+1+1 3+3+2 3+3+1+1
4	odd	0	
4	even	14	8+2 8+1+1 4+4+2 4+4+1+1 4+2+2+2 4+2+2+1+1 4+2+1+1+1+1 4+1+1+1+1+1+1 2+2+2+2+2 2+2+2+2+1+1 2+2+2+1+1+1+1 2+2+1+1+1+1+1+1 2+1+1+1+1+1+1+1+1 1+1+1+1+1+1+1+1+1

5	odd	0	
5	even	4	10+2 10+1+1 5+5+2 5+5+1+1
6	odd	0	
6	even	8	12+2 12+1+1 6+6+2 6+6+1+1 6+3+3+2 6+3+3+1+1 3+3+3+3+2 3+3+3+3+1+1
7	odd	0	
7	even	4	14+2 14+1+1 7+7+2 7+7+1+1

Graph

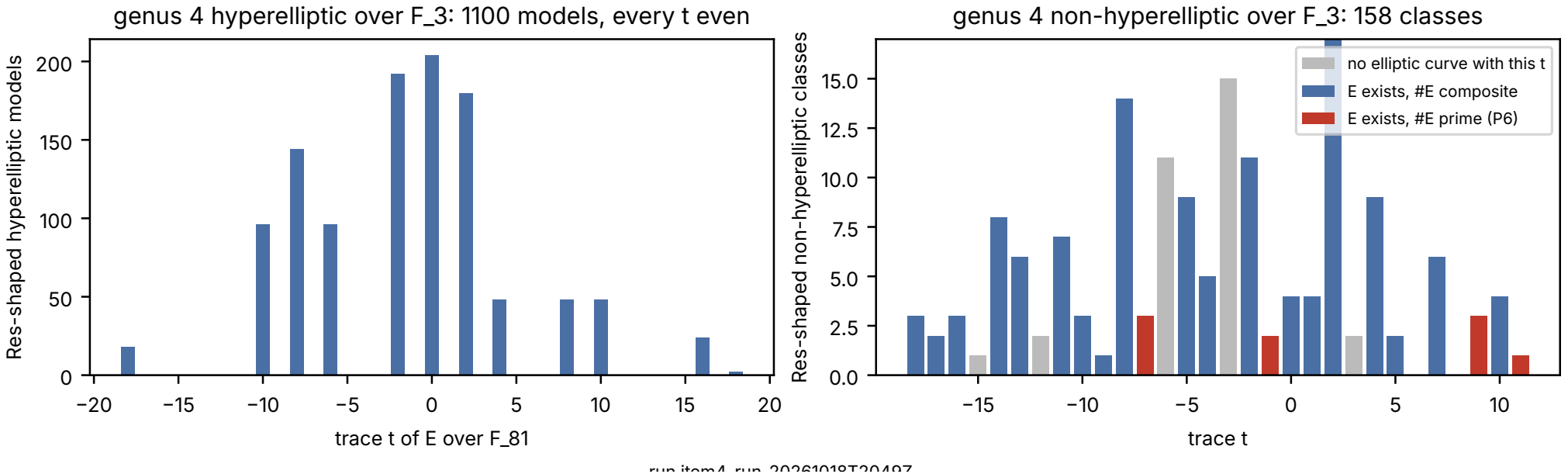


Figure 1: Res-shaped genus-4 curves over F_3 by trace: hyperelliptic models (left), non-hyperelliptic classes (right); exact census.